Assignment 7: Business Intelligence and Data Visualisation with Tableau

Dataset: World Bank population by country and year (SP.POP.TOTL, 1960 to 2023). Workbook: Population_BI.twbx.

1. Benefits of Business Intelligence

• **Faster, better decisions.** A 2024 study of BI adoption reports better decision speed, accuracy and competitiveness (SAGE Open, doi:10.1177/02663821241289809).

• **One version of the truth.** Dashboards read from one governed source, so departments stop arguing over conflicting spreadsheets (BARC, benefits of BI).

• **Operational efficiency.** Reports that took days can be refreshed on demand, and staff spend less time assembling data (BARC; Park University).

• **Better customer and market insight.** Segmentation and trend views show where demand is growing (Databricks, 'How Business Intelligence Drives Smart Decision-Making', 2025).

• **Data quality and accountability.** Shared definitions and visible metrics make errors easier to spot and owners easier to name (BARC).

Sources were found by web search. The summaries are from search results and were not checked against each full article.

2. Setup and data connection

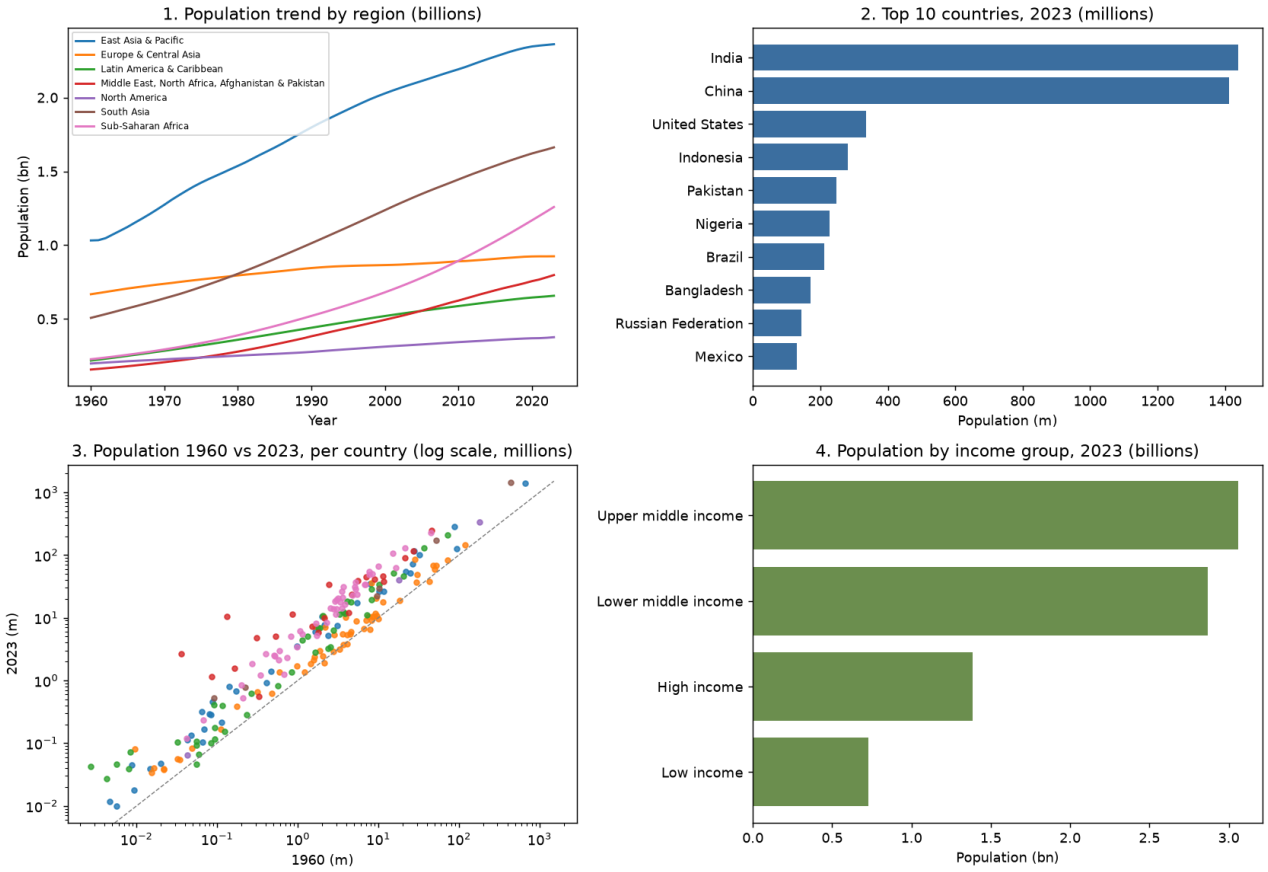
Tableau Desktop is not installed on the build machine, and no Redshift cluster is available. The data was therefore connected as a text file (Route A), packaged in the .twbx with the CSV. The Redshift route (Route B: COPY from S3, then a Tableau Redshift connection on port 5439 with SSL) is written out in *docs/connection_guide.md* but was not run. The CSV has 13,858 rows: 217 countries and territories, 1960 to 2023. The 79 World Bank aggregates (World, regions, income groups) are removed using the country metadata.

3. Worksheets, operators, functions, sorting and filtering

Worksheet	Chart	Operations
Population trend by region	Line	Year range filter; SUM(population); colour by region
Top 10 countries, latest year	Bar, sorted	Filter year = 2023 and pop_rank_in_year 1 to 10; sort by population, descending
Population 1960 vs latest	Scatter	Calculated fields IF year = 1960 / 2023 THEN population END; detail on country
Population by income group, latest	Bar, sorted	Filter year = 2023; sort descending
Population dashboard	4 sheets	Shared region and year filters

Rank within each year is computed in the data preparation step, so the Top 10 filter does not depend on a table calculation.

4. Charts and insights



Preview drawn with matplotlib from the same data; not a Tableau screenshot.

• World population rose from 3.01 billion in 1960 to 8.04 billion in 2023, an increase of 167%.

• Growth was uneven. Sub-Saharan Africa grew 453% and the Middle East and North Africa group 405%, while Europe and Central Asia grew 38%. The line chart shows the regions on separate paths over the whole period.

• In 2023, India (1.44 bn) and China (1.41 bn) were the two largest countries. The next are the United States (337 m) and Indonesia (281 m).

• Upper middle income countries held 3.06 bn people, more than the high income group at 1.38 bn.

• The scatter plot points above the dashed line grew. The steepest multiples are in small states such as the United Arab Emirates and Qatar, where a small 1960 base inflates the ratio. Read these with the absolute values, not the ratio alone.

5. Limitations

• The workbook was written by hand as Tableau XML and has not been opened in Tableau Desktop. Section 6 of docs/connection_guide.md gives the rebuild steps if it fails.

• The Redshift route was not run. The CSV route is the one that was tested.

• Screenshots of the connection steps and the Tableau charts are still to be added after the workbook opens.

• The presentation is this report. No separate slide deck was made.

6. References

• World Bank Open Data, Population, total (SP.POP.TOTL). https://data.worldbank.org/indicator/SP.POP.TOTL

• World Bank Countries and Regions API. https://api.worldbank.org/v2/country

• BARC. The Benefits of Business Intelligence. https://barc.com/benefits-business-intelligence/

• SAGE Open (2024). The impact of business intelligence on promoting decision-making. doi:10.1177/02663821241289809

• Databricks (2025). How Business Intelligence Drives Smart Decision-Making. https://www.databricks.com/blog/how-business-intelligence-drives-smart-decision-making

• Harvard Business School Online. The Advantages of Data-Driven Decision-Making. https://online.hbs.edu/blog/post/data-driven-decision-making

• Tableau Help: connecting to text files and to Amazon Redshift (Tableau documentation).